SUPPLY CHAIN CONTROL TOWER PROJECT

Objective:
Build an end-to-end Supply Chain Control Tower using Excel, SQL, and Power BI to analyze demand, inventory, suppliers, forecasting, warehouses, and transportation, and provide business recommendations to improve service level, inventory performance, and supply chain efficiency.

Project Summary

Business Problem

Design a Supply Chain Control Tower to monitor operational performance across demand planning, inventory management, transportation, supplier performance, warehouse operations, forecasting, and order fulfillment.

Tools Used

- Excel
- SQL (SQLite)
- Power BI

Dataset

- 25,000 Orders
- 8 Business Tables
- 5,000 Shipments
- 8 Warehouses
- 498 SKUs

Deliverables

- 7 Interactive Dashboards
- 30+ DAX Measures
- 40+ SQL Validation Queries
- 6 Business Investigations
- Executive Recommendations

DATASETS USED

1. Orders
2. Product
3. Inventory

4. Supplier
5. Supplier_Product_Map
6. Transportation
7. Warehouse
8. Forecast

PHASE 1: EXCEL ANALYSIS

Purpose:
Understand the business, clean data, calculate KPIs, identify patterns, and develop initial insights.

Analysis Completed:

1. Warehouse Demand Analysis
   Question:
   Which warehouse has the highest demand?

Measure:
Demand_Qty

Finding:
WH5 had the highest demand.

Business Impact:
Identify high-demand locations and review inventory capacity, replenishment strategy, and warehouse expansion opportunities.

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2. Delayed Orders Analysis

Question:
Which warehouse has the highest delayed orders?

Measure:
Count of delayed orders

Finding:
WH7 had the highest delayed orders.

Business Impact:
Investigate warehouse processes, supplier lead times, and transportation performance.

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3. Regional Demand Analysis

Question:
Which region generates the highest demand?

Measure:
Demand_Qty

Finding:
North region generated the highest demand.

Business Impact:
Improve regional inventory positioning and demand planning.

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4. Product Category Analysis

Question:
Which category generates the highest demand?

Measure:
Demand_Qty

Finding:
Parts category had the highest demand.

Business Impact:
Focus planning efforts on critical product categories.

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5. Fill Rate Analysis

Question:
How effectively are customer demands being fulfilled?

Measure:
Delivered_Qty / Demand_Qty

Finding:
Overall Fill Rate was 66.9%, later confirmed in the Power BI Order Fulfillment dashboard.

Business Impact:
Measure service level and identify supply chain performance gaps.

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6. Inventory Risk Analysis

Question:
Which SKUs are at inventory risk?

Measures:
Current Stock
Safety Stock
Reorder Point

Business Impact:
Prevent stockouts and improve inventory planning.

PHASE 2: SQL ANALYSIS

Purpose

Investigate root causes by connecting multiple datasets and answering business questions that cannot be easily solved through Excel alone.

The objective of this phase is to develop data-driven supply chain insights by combining demand, inventory, supplier, transportation, warehouse, and forecasting data using SQL.

Investigation 1: Supplier Dependency Analysis

Business Question

Which suppliers support the highest demand volume across the network?

Business Purpose

Identify supplier concentration risk and determine whether the business is heavily dependent on a limited number of suppliers.

Tables Used

Orders

Columns Used:

- SKU_ID
- Demand_Qty

Purpose:

- Provides customer demand volume.

Supplier_Product_Map

Columns Used:

- SKU_ID
- Supplier_ID

Purpose:

- Maps products to suppliers.

Supplier

Columns Used:

- Supplier_ID
- Supplier_Name

Purpose:

- Provides supplier master data.

Data Relationship

Orders
↓ SKU_ID
Supplier_Product_Map
↓ Supplier_ID
Supplier

Analytical Approach

Measure

Demand_Qty

Dimension

Supplier_Name

Aggregation

SUM(Demand_Qty)

Reason

Determine total demand support by each supplier.

SQL Query

```
SELECT
    s.Supplier_Name,
    SUM(o.Demand_Qty) AS Total_Demand
FROM Orders o
JOIN Supplier_Product_Map spm
ON o.SKU_ID = spm.SKU_ID
JOIN Supplier s
ON spm.Supplier_ID = s.Supplier_ID
GROUP BY s.Supplier_Name
ORDER BY Total_Demand DESC
LIMIT 3;
```

Result

	Supplier_Name	Total_Demand
1	Supplier_38	143714
2	Supplier_49	134380
3	Supplier_14	129265

Findings

Supplier_38 supports the highest demand volume (143,714 units), followed by Supplier_49 (134,380 units) and Supplier_14 (129,265 units).

The top suppliers collectively support a significant portion of total network demand.

Business Impact

The business demonstrates supplier concentration risk, where a significant portion of total demand is supported by a limited number of suppliers.

Any disruption affecting these suppliers could negatively impact product availability, fill rate, and customer service performance.

Recommendations

- Review supplier capacity and resilience for the top-demand suppliers.
- Monitor supplier concentration risk regularly.
- Evaluate alternate sourcing opportunities for critical SKUs.
- Review inventory strategies for highly dependent supplier-product combinations.

Key Supply Chain Concepts Learned

- Supplier Dependency Analysis
- Demand Aggregation
- Supplier Concentration Risk
- Multi-Table JOINs
- Business Measures and Dimensions
- Aggregation using SUM()
- GROUP BY Analysis

Interview Talking Point

Used SQL to connect demand and supplier master data through a mapping table to identify supplier dependency risks.

Identified the top demand-supporting suppliers and recommended supplier diversification and risk mitigation strategies.

Investigation 1A: Supplier Performance Risk Analysis

Business Question

Among the highest-demand suppliers, which suppliers present the greatest operational risk based on delivery performance, lead time, and quality metrics?

Business Purpose

Determine whether critical suppliers can reliably support customer demand and identify potential risks to service levels.

Tables Used

Orders

Columns Used:

- SKU_ID
- Demand_Qty

Supplier_Product_Map

Columns Used:

- SKU_ID
- Supplier_ID

Supplier

Columns Used:

- Supplier_ID
- Supplier_Name
- OTD_Pct
- Lead_Time_Days
- Defect_Rate

Data Relationship

```
Orders
   ↓ SKU_ID
Supplier_Product_Map
   ↓ Supplier_ID
Supplier
```

Analytical Approach

Measure

Demand_Qty

Aggregation

SUM(Demand_Qty)

Supplier KPIs

- OTD_Pct

- Lead_Time_Days
- Defect_Rate

Purpose

Evaluate supplier reliability, responsiveness, and quality.

SQL Query

```
SELECT
    s.Supplier_Name,
    SUM(o.Demand_Qty) AS Total_Demand,
    s.OTD_Pct,
    s.Lead_Time_Days,
    s.Defect_Rate
FROM Orders o
JOIN Supplier_Product_Map spm
ON o.SKU_ID = spm.SKU_ID
JOIN Supplier s
ON spm.Supplier_ID = s.Supplier_ID
GROUP BY
    s.Supplier_Name,
    s.OTD_Pct,
    s.Lead_Time_Days,
    s.Defect_Rate
ORDER BY Total_Demand DESC
LIMIT 3;
```

Result

	Supplier_Name	Total_Demand	OTD_Pct	Lead_Time_Days	Defect_Rate
1	Supplier_38	143714	65	14	6.06
2	Supplier_49	134380	75	5	1.27
3	Supplier_14	129265	70	3	7.31

Findings

Supplier_38

- Highest demand contributor.
- Lowest OTD performance (65%).
- Longest lead time (14 days).
- Elevated defect rate (6.06%).

Assessment:

Highest operational risk among top-demand suppliers.

Supplier_49

- Second-highest demand contributor.
- Highest OTD performance (75%).
- Lowest defect rate (1.27%).
- Moderate lead time (5 days).

Assessment:

Strongest performing supplier among the top-demand contributors.

Supplier_14

- Third-highest demand contributor.
- Shortest lead time (3 days).
- Highest defect rate (7.31%).

Assessment:

Quality-related supply chain risk.

Business Impact

The organization is highly dependent on suppliers exhibiting relatively weak delivery reliability and quality performance.

Supplier_38 represents the most significant supply chain risk due to its combination of:

- High demand dependency
- Low OTD performance
- Long lead time
- Elevated defect rate

Any disruption could negatively impact inventory availability, fill rate, and customer service levels.

Recommendations

Immediate Actions

- Conduct supplier performance review with Supplier_38.
- Investigate root causes of low OTD performance.

- Review lead-time reduction opportunities.
- Investigate quality issues affecting Supplier_14.

Medium-Term Actions

- Evaluate shifting demand toward higher-performing suppliers where feasible.
- Develop alternate sourcing strategies for critical SKUs.
- Review safety stock levels for products supplied by Supplier_38.
- Implement supplier performance monitoring dashboards.

Key Supply Chain Concepts Learned

- Supplier Performance Management
- OTD Analysis
- Lead Time Analysis
- Quality Risk Assessment
- Demand Dependency Risk
- Supplier Segmentation
- Data-Driven Decision Making
- Supply Chain Risk Management

Interview Talking Point

Extended supplier dependency analysis by integrating supplier performance KPIs.

Identified that the highest-demand supplier also exhibited the weakest delivery reliability and longest lead time, creating significant operational risk.

Recommended supplier development initiatives, sourcing mitigation strategies, and inventory risk controls.

Investigation 2: Inventory Risk Analysis

Business Question

Which high-demand SKUs are operating below safety stock levels?

Business Purpose

Identify inventory shortage risks by combining customer demand and inventory position data. The objective is to proactively identify products that may cause stockouts, service-level failures, and customer fulfillment issues.

Tables Used

Orders

Columns Used:

- SKU_ID
- Demand_Qty

Purpose:

- Provides customer demand volume for each product.
- Helps identify business-critical SKUs.

Inventory

Columns Used:

- SKU_ID
- Current_Stock
- Safety_Stock

Purpose:

- Provides inventory availability and safety stock requirements.
- Enables identification of inventory shortages.

Data Relationship

```
Orders
   ↓ SKU_ID
Inventory
```

Analytical Challenge

The Orders and Inventory tables operate at different levels of detail (grain).

Orders Table Grain

One Row = One Order Transaction

Inventory Table Grain

One Row = One SKU in One Warehouse

Desired Output Grain

One Row = One SKU

To avoid duplicate demand calculations, both datasets were aggregated to SKU level before joining.

Analytical Approach

Step 1: Aggregate Demand

Measure

- Demand_Qty

Aggregation

- SUM(Demand_Qty)

Purpose

- Calculate total customer demand for each SKU.

Step 2: Aggregate Inventory

Measures

- Current_Stock
- Safety_Stock

Aggregation

- SUM(Current_Stock)
- SUM(Safety_Stock)

Purpose

- Calculate total inventory position across the network for each SKU.

Step 3: Calculate Inventory Gap

Formula:

Inventory_Gap =
Total_Current_Stock - Total_Safety_Stock

Purpose

Measure inventory shortage or surplus relative to safety stock requirements.

Step 4: Identify Inventory Risk

Filter:

Total_Current_Stock < Total_Safety_Stock

Purpose:

Identify SKUs operating below safety stock levels.

SQL Query

```
SELECT
    o.SKU_ID,
    o.Total_Demand,
    i.Total_Current_Stock,
    i.Total_Safety_Stock,
    (i.Total_Current_Stock - i.Total_Safety_Stock) AS Inventory_Gap

FROM
(
    SELECT
        SKU_ID,
        SUM(Demand_Qty) AS Total_Demand
    FROM Orders
    GROUP BY SKU_ID
) o

LEFT JOIN

(
    SELECT
        SKU_ID,
        SUM(Current_Stock) AS Total_Current_Stock,
        SUM(Safety_Stock) AS Total_Safety_Stock
    FROM Inventory
    GROUP BY SKU_ID
) i

ON o.SKU_ID = i.SKU_ID

WHERE i.Total_Current_Stock < i.Total_Safety_Stock

ORDER BY o.Total_Demand DESC;
```

Result

	SKU_ID	Total_Demand	Total_Current_Stock	Total_Safety_Stock	Inventory_Gap
1	SKU0476	9349	185	336	-151
2	SKU0107	8906	1	51	-50

Findings

SKU0476

- Highest demand among inventory-risk SKUs.
- Current stock of 185 units versus safety stock requirement of 336 units.
- Inventory deficit of 151 units.

Assessment

Highest inventory exposure within the network.

SKU0107

- Demand volume of 8,906 units.
- Current inventory reduced to only 1 unit.
- Inventory deficit of 50 units.

Assessment

Immediate stockout risk due to critically low inventory availability.

Network-Level Observation

- Inventory table contains 498 unique SKUs.
- Only 2 SKUs were identified below safety stock levels.

Inventory Health Rate:

$(496 / 498) \times 100$
$= 99.6\%$

Assessment

Overall inventory health across the network is strong, with only localized inventory shortages requiring attention.

Business Impact

The overall inventory network appears healthy, with approximately 99.6% of SKUs operating at or above safety stock levels.

However:

- SKU0476 represents a significant inventory exposure due to high demand and a substantial inventory deficit.
- SKU0107 faces immediate stockout risk because inventory availability is nearly depleted.

Failure to replenish these products could negatively impact:

- Fill Rate
- Service Level
- Customer Satisfaction
- Revenue Performance

Recommendations

Immediate Actions

- Prioritize replenishment for SKU0107.
- Review inventory planning parameters for SKU0476.
- Expedite replenishment orders for both SKUs.

Medium-Term Actions

- Review safety stock policies for critical products.
- Implement automated inventory-risk alerts.
- Improve inventory monitoring through Control Tower dashboards.
- Conduct periodic inventory health reviews for high-demand SKUs.

Key Supply Chain Concepts Learned

- Inventory Risk Analysis
- Safety Stock Management
- Inventory Gap Analysis
- Network Inventory Visibility
- Inventory Health Assessment
- Demand Prioritization
- Data Grain Analysis
- SQL Subqueries
- LEFT JOIN Strategy
- Pre-Aggregation Techniques

Interview Talking Point

Used SQL subqueries to aggregate demand and inventory data at SKU level, eliminating data duplication caused by different table grains.

Calculated inventory gaps and identified high-demand SKUs operating below safety stock levels. Discovered that only 0.4% of SKUs were at inventory risk, while highlighting two critical products requiring immediate replenishment actions.

Investigation 3: Forecast Accuracy Analysis

Business Question

Which SKUs have the highest forecast error?

Business Purpose

Identify products with poor forecast accuracy to improve demand planning, reduce stockouts, minimize excess inventory, and optimize inventory investment.

Tables Used

Forecast

Columns Used

- SKU_ID
- Forecast_Qty
- Actual_Demand

Purpose

- Forecast_Qty represents planned customer demand.
- Actual_Demand represents realized customer demand.
- Enables measurement of planning accuracy.

Data Relationship

Only a single table is required.

Forecast

Analytical Approach

Desired Output Grain

One Row = One SKU

The Forecast table contains multiple records by SKU, warehouse, and month. Therefore, the data is aggregated to SKU level to evaluate overall planning accuracy for each product.

Measure 1

Forecast_Qty

Aggregation

SUM(Forecast_Qty)

Purpose

Calculate the total planned demand for each SKU.

Measure 2

Actual_Demand

Aggregation

SUM(Actual_Demand)

Purpose

Calculate the total actual customer demand for each SKU.

Measure 3

Forecast Error

Formula

Forecast Error =
Total_Actual_Demand - Total_Forecast

Purpose

Determine whether demand was:

- Higher than forecast (Underforecast)
- Lower than forecast (Overforecast)

Interpretation

Positive Forecast Error

Actual > Forecast

Business Meaning

Underforecasting

Negative Forecast Error

Forecast > Actual

Business Meaning

Overforecasting

Measure 4

Absolute Forecast Error

Formula

ABS(Forecast Error)

Purpose

Measures the magnitude of forecasting error regardless of direction and enables ranking of the largest planning deviations.

SQL Query

```
SELECT
    SKU_ID,
    SUM(Forecast_Qty) AS Total_Forecast,
    SUM(Actual_Demand) AS Total_Actual_Demand,
    SUM(Actual_Demand) - SUM(Forecast_Qty) AS Forecast_Error,
    ABS(SUM(Actual_Demand) - SUM(Forecast_Qty)) AS Absolute_Forecast_Error
FROM Forecast
GROUP BY SKU_ID
ORDER BY Absolute_Forecast_Error DESC
LIMIT 5;
```

Result

	SKU_ID	Total_Actual_Demand	Total_Forecast	Forecast_Error	Absolute_Forecast_Error
1	SKU0308	4643	2762	1881	1881
2	SKU0186	2431	4221	-1790	1790
3	SKU0235	5061	3324	1737	1737
4	SKU0472	4606	6250	-1644	1644
5	SKU0464	4479	6099	-1620	1620

Findings

SKU0308

- Largest forecast deviation in the network.

- Demand exceeded forecast by **1,881 units**.

Assessment

Highest underforecast risk, increasing the likelihood of stockouts and service-level failures.

SKU0235

- Demand exceeded forecast by **1,737 units**.

Assessment

Additional underforecasting issue requiring review of planning assumptions.

SKU0186

- Forecast exceeded actual demand by **1,790 units**.

Assessment

Largest overforecasting case, indicating excess inventory exposure.

SKU0472

- Forecast exceeded actual demand by **1,644 units**.

Assessment

Potential inventory carrying cost and warehouse utilization concerns.

SKU0464

- Forecast exceeded actual demand by **1,620 units**.

Assessment

Significant planning deviation results in excess inventory risk.

Business Impact

The analysis identified both underforecasting and overforecasting across critical SKUs.

Underforecasting may lead to:

- Stockouts
- Reduced service level
- Lost sales
- Emergency replenishment

Overforecasting may lead to:

- Excess inventory
- Higher inventory carrying costs
- Poor warehouse space utilization
- Working capital tied up in inventory

The presence of significant forecast deviations in both directions indicates opportunities to improve forecasting accuracy and demand planning processes.

Recommendations

Immediate Actions

- Review planning assumptions for the top forecast-error SKUs.
- Investigate root causes behind significant forecasting deviations.
- Adjust future forecasts using historical forecast performance.

Medium-Term Actions

- Establish forecast accuracy as a recurring planning KPI.
- Improve collaboration between Sales, Demand Planning, and Supply Planning.
- Incorporate historical forecast error into future planning cycles.
- Develop Power BI dashboards to monitor forecast accuracy and forecast bias over time.

Key Supply Chain Concepts Learned

- Forecast Accuracy Analysis
- Demand Planning
- Forecast Error
- Absolute Forecast Error
- Underforecast vs Overforecast
- Planning Performance Measurement
- Aggregate Functions (SUM)
- ABS() Function
- GROUP BY Analysis
- Planning KPI Development

Interview Talking Point

Used SQL to evaluate demand planning performance by comparing forecasted demand with actual customer demand at the SKU level. Calculated forecast error and absolute forecast error to identify products with the largest planning deviations, highlighting both stockout risks caused by underforecasting and excess inventory risks caused by overforecasting. Recommended improving forecast accuracy through historical error analysis, cross-functional planning, and continuous KPI monitoring.

Investigation 4: Warehouse Inventory Risk Concentration Analysis

Business Question

Which warehouses contain the highest concentration of inventory-risk SKUs while supporting high customer demand?

Business Purpose

Identify warehouses that contain the greatest number of inventory-risk SKUs while supporting significant customer demand. This enables the supply chain team to prioritize warehouse-level replenishment efforts and reduce the risk of stockouts affecting customer service.

Tables Used

Orders

Columns Used

- Warehouse_ID
- Demand_Qty

Purpose

Provides customer demand handled by each warehouse.

Inventory

Columns Used

- Warehouse_ID
- SKU_ID
- Current_Stock
- Safety_Stock

Purpose

Identifies inventory-risk SKUs by comparing current inventory against safety stock.

Data Relationship

Orders

↓ Warehouse_ID

Inventory

Table Grain

Orders

One row represents **one order transaction.**

Inventory

One row represents **one SKU in one Warehouse.**

Desired Output Grain

One row represents **one Warehouse.**

Analytical Approach

Measure 1

Demand_Qty

Aggregation

SUM(Demand_Qty)

Purpose

Calculate the total customer demand handled by each warehouse.

Measure 2

Risk SKU

Business Rule

Current_Stock < Safety_Stock

Aggregation

COUNT(SKU_ID)

Purpose

Count the number of inventory-risk SKUs within each warehouse.

SQL Query

```sql
SELECT
    o.Warehouse_ID,
    o.Total_Demand,
    COALESCE(i.Risk_SKU_Count, 0) AS Risk_SKU_Count

FROM
(
    SELECT
        Warehouse_ID,
        SUM(Demand_Qty) AS Total_Demand
    FROM Orders
    GROUP BY Warehouse_ID
) o

LEFT JOIN

(
    SELECT
        Warehouse_ID,
        COUNT(SKU_ID) AS Risk_SKU_Count
    FROM Inventory
    WHERE Current_Stock < Safety_Stock
    GROUP BY Warehouse_ID
) i

ON o.Warehouse_ID = i.Warehouse_ID

ORDER BY
    o.Total_Demand DESC;
```

Result

	Warehouse_ID	Total_Demand	Risk_SKU_Count
1	WH5	482302	45
2	WH7	476289	53
3	WH3	470550	43
4	WH8	468281	38
5	WH4	466856	43
6	WH1	466039	37
7	WH6	462417	30

Findings

- Warehouses were ranked by total customer demand alongside the number of inventory-risk SKUs.
- WH7 had the highest Risk SKU Count (53), while WH5 processed the highest demand.
- The analysis enables planners to identify operational hotspots where inventory shortages are concentrated despite high warehouse activity.

Business Impact

Traditional warehouse-level inventory aggregation indicated that all warehouses held inventory above their overall safety stock levels. However, this masked SKU-level shortages within individual warehouses.

By analyzing the concentration of inventory-risk SKUs instead of total warehouse inventory, the Control Tower provides a more accurate representation of operational inventory risk. This allows planners to prioritize replenishment efforts toward warehouses where multiple critical SKUs are below safety stock, improving inventory visibility and reducing potential stockouts.

Recommendations

- Prioritize replenishment planning for warehouses with the highest number of inventory-risk SKUs.
- Review inventory allocation policies to reduce SKU-level shortages within warehouses.
- Investigate recurring inventory-risk patterns to identify root causes such as inaccurate forecasting, replenishment delays, or supplier issues.
- Develop warehouse-level dashboards to continuously monitor inventory-risk concentration.

Key Supply Chain Concepts Learned

- Warehouse Inventory Risk

- Inventory Visibility
- Safety Stock Monitoring
- Inventory Concentration Risk
- Warehouse Prioritization
- Business Rules using SQL (WHERE)
- Aggregation using COUNT()
- Multi-table Analysis using LEFT JOIN
- Data-driven Warehouse Planning

Interview Talking Point

Designed a warehouse-level inventory risk analysis by integrating demand and inventory data. Instead of relying on overall warehouse inventory balances, I identified warehouses with the highest concentration of SKUs below safety stock while considering customer demand. This provided a more actionable view of inventory risk and supported warehouse prioritization for replenishment planning.

Investigation 5: Order Fulfillment Performance Analysis

Business Question

What percentage of customer orders are fulfilled with the complete quantity demanded?

Business Purpose

Evaluate customer order Fulfillment performance by measuring the percentage of orders where the delivered quantity meets or exceeds the quantity demanded.

The analysis helps identify service-level gaps caused by incomplete order Fulfillment and provides a foundation for investigating potential issues related to inventory availability, forecasting, warehouse operations, and supplier performance.

Tables Used

Orders

Columns Used:

- Order_ID
- Demand_Qty
- Delivered_Qty

Purpose:

- Order_ID identifies individual customer orders.
- Demand_Qty represents the quantity requested by the customer.
- Delivered_Qty represents the quantity actually delivered.

Table Grain

One row represents one customer order transaction.

Desired Output Grain

One row representing overall network-level In-Full order Fulfillment performance.

Analytical Approach

Measure 1: Total Orders

Column:

Order_ID

Aggregation:

COUNT(Order_ID)

Purpose:

Calculate the total number of customer orders evaluated.

Measure 2: In-Full Orders

Business Rule:

Delivered_Qty >= Demand_Qty

An order is classified as **In Full** when the delivered quantity is equal to or greater than the quantity demanded.

Aggregation:

```
SUM(

    CASE

        WHEN Delivered_Qty >= Demand_Qty THEN 1

        ELSE 0

    END

)
```

Purpose:

Count the number of customer orders that were fulfilled in full.

The CASE WHEN statement evaluates each order individually:

- 1 = In Full
- 0 = Not In Full

SUM() then adds all the orders classified as 1 to calculate the total number of In-Full orders.

Measure 3: In-Full Rate

Calculation:

In-Full Rate =

(In-Full Orders / Total Orders) × 100

Purpose:

Measure the percentage of customer orders where the complete requested quantity was delivered.

SQL Query

```
SELECT
    COUNT(Order_ID) AS Total_Orders,

    SUM(
        CASE
            WHEN Delivered_Qty >= Demand_Qty THEN 1
            ELSE 0
        END
    ) AS In_Full_Orders,

    ROUND(
        100.0 *
        SUM(
            CASE
                WHEN Delivered_Qty >= Demand_Qty THEN 1
                ELSE 0
```

```
            END
        )
        / COUNT(Order_ID),
        2
    ) AS In_Full_Rate_Pct

FROM Orders;
```

Result

	Total_Orders	In_Full_Orders	In_Full_Rate_Pct
1	25000	12587	50.35

Additional calculation:

- **Not-In-Full Orders:** 12,413
- **Not-In-Full Rate:** 49.65%

Findings

Out of **25,000 customer orders**, **12,587 orders were fulfilled in full**, resulting in an overall **In-Full Rate of 50.35%.**

Approximately **49.65% of customer orders did not receive the complete quantity demanded**, indicating a significant gap in order Fulfillment performance.

This analysis measures the **In-Full component of customer service performance**. The result suggests that incomplete order Fulfillment requires further investigation to determine whether the issue is associated with inventory shortages, forecast inaccuracies, warehouse-level inventory availability, supplier constraints, or other operational factors.

OTIF Data Limitation

The original objective was to evaluate **OTIF (On Time In Full)** performance.

However, true OTIF cannot be calculated using the currently available dataset.

The dataset provides:

- Demand_Qty
- Delivered_Qty

which allows calculation of the **In-Full** component.

However, the dataset does not contain sufficient order-level information such as:

- Promised or Required Delivery Date
- Actual Delivery Date

Therefore, the **On-Time** component cannot be reliably determined.

The Transportation table contains Delivery_Days, but without a defined committed delivery time and a direct Order-to-Shipment relationship, it cannot be used to calculate true order-level OTIF.

Therefore, the analysis focuses on **In-Full performance rather than presenting an inaccurate OTIF KPI.**

Business Impact

An In-Full Rate of **50.35%** indicates that nearly half of customer orders were not fulfilled with the complete quantity requested.

Poor In-Full performance can potentially contribute to:

- Reduced customer service levels.
- Increased backorders or partial deliveries.
- Lower customer satisfaction.
- Additional transportation or handling requirements for remaining quantities.
- Potential loss of future customer demand.

The result highlights the need for deeper analysis to identify where Not-In-Full orders are concentrated and determine the underlying operational causes.

Recommendations

Immediate Actions

- Identify the SKUs contributing most frequently to Not-In-Full orders.
- Investigate whether Not-In-Full orders are concentrated within specific warehouses.
- Review inventory availability for frequently under-fulfilled SKUs.
- Compare under-fulfilled SKUs against inventory-risk SKUs identified in Investigation 2.

Further Analysis

- Analyse In-Full performance by warehouse.
- Analyse In-Full performance by customer region.
- Analyse In-Full performance by SKU and product category.
- Investigate relationships between forecast errors and under-fulfilled orders.

- Investigate whether warehouses with higher Risk SKU Counts also experience lower In-Full performance.

These analyses have been incorporated into the Power BI dashboards where applicable.

Key Supply Chain Concepts Learned

- Order Fulfillment
- In-Full Performance
- Service-Level Measurement
- Demand vs Delivered Quantity
- OTIF Concept and Data Requirements
- Conditional Aggregation
- SQL CASE WHEN
- SUM(CASE WHEN...)
- Percentage KPI Calculation
- Data Limitation Identification

Interview Talking Point

Used SQL to evaluate order Fulfillment performance across 25,000 customer orders by comparing demanded and delivered quantities. Applied conditional aggregation using CASE WHEN to classify and count In-Full orders and calculated an overall In-Full Rate of 50.35%. I also evaluated whether the dataset supported OTIF analysis and identified that promised and actual delivery-date information was unavailable, so I avoided presenting an inaccurate OTIF metric and focused on the measurable In-Full component.

Investigation 6: Transportation Carrier Performance Analysis

Business Question

How do transportation carriers perform in terms of delivery speed and freight cost, and are there route-specific differences that should influence carrier selection?

Business Purpose

Evaluate transportation carrier performance by comparing shipment volume, average delivery time, and freight cost.

The objective is to identify potential cost-service trade-offs between carriers and determine whether carrier performance varies across different transportation lanes.

This analysis supports more informed carrier allocation decisions based on shipment urgency, transportation cost, and route-specific performance.

Tables Used

Transportation

Columns Used

- Shipment_ID
- Carrier
- Origin
- Destination
- Freight_Cost
- Delivery_Days

Purpose

- Shipment_ID identifies individual shipments.
- Carrier identifies the transportation service provider.
- Origin identifies the shipment starting location.
- Destination identifies the shipment delivery location.
- Freight_Cost represents transportation cost.
- Delivery_Days represents the number of days required for delivery.

Table Grain

One row represents one shipment transaction.

Analysis 1: Overall Carrier Performance

Business Question

How do carriers compare the overall transportation network in terms of shipment volume, average delivery time, and freight cost?

Desired Output Grain

One row represents one carrier.

Analytical Approach

Measure 1: Total Shipments

Aggregation:

COUNT(Shipment_ID)

Purpose:

Measure the number of shipments handled by each carrier.

Measure 2: Average Delivery Days

Aggregation:

AVG(Delivery_Days)

Purpose:

Measure the average delivery speed of each carrier.

Lower average delivery days indicate faster transportation performance.

Measure 3: Average Freight Cost

Aggregation:

AVG(Freight_Cost)

Purpose:

Compare the average freight cost associated with each carrier.

Measure 4: Total Freight Cost

Aggregation:

SUM(Freight_Cost)

Purpose:

Measure the total transportation spend associated with each carrier.

SQL Query – Overall Carrier Performance

```
SELECT

    Carrier,

    COUNT(Shipment_ID) AS Total_Shipments,

    ROUND(AVG(Delivery_Days), 2) AS Avg_Delivery_Days,

    ROUND(AVG(Freight_Cost), 2) AS Avg_Freight_Cost,

    ROUND(SUM(Freight_Cost), 2) AS Total_Freight_Cost
```

```sql
FROM Transportation
GROUP BY Carrier
ORDER BY Avg_Delivery_Days ASC;
```

Result

Carrier	Total Shipments	Avg Delivery Days	Avg Freight Cost	Total Freight Cost
UPS	1,268	4.87	10,379.64	13,161,379
FedEx	1,234	4.88	10,194.05	12,579,455
BlueDart	1,234	4.90	10,587.59	13,065,081
DHL	1,264	4.93	10,476.71	13,242,565

Initial Findings

Overall carrier delivery performance was relatively similar across the transportation network.

UPS recorded the lowest average delivery time at **4.87 days**, while DHL recorded the highest at **4.93 days**, representing a difference of only **0.06 days**.

FedEx recorded the lowest average freight cost at **10,194.05**, while BlueDart recorded the highest at **10,587.59**.

Since network-level averages showed limited differentiation in delivery performance, further analysis was conducted at the transportation-lane level to determine whether route-specific performance differences were being hidden by overall averages.

Analysis 2: Transportation Lane Performance

Business Question

Does carrier cost and delivery performance vary across specific Origin–Destination transportation lanes?

Business Purpose

Analyse carrier performance at the transportation-lane level to identify route-specific cost-service trade-offs that may not be visible in overall carrier averages.

Desired Output Grain

One row represents one:

Origin + Destination + Carrier combination

SQL Query – Lane-Level Carrier Performance

```
SELECT
    Origin,
    Destination,
    Carrier,
    COUNT(Shipment_ID) AS Total_Shipments,
    ROUND(AVG(Delivery_Days), 2) AS Avg_Delivery_Days,
    ROUND(AVG(Freight_Cost), 2) AS Avg_Freight_Cost
FROM Transportation
GROUP BY
    Origin,
    Destination,
    Carrier
ORDER BY
    Origin,
    Destination,
    Avg_Delivery_Days ASC;
```

Result

	Origin	Destination	Carrier	Total_Shipments	Avg_Delivery_Days	Avg_Freight_Cost
1	Chennai	Bangalore	BlueDart	432	4.87	10934.81
2	Chennai	Bangalore	UPS	422	4.87	10511.77
3	Chennai	Bangalore	DHL	390	5.02	10615.68
4	Chennai	Bangalore	FedEx	410	5.07	10073.53
5	Mumbai	Delhi	DHL	425	4.83	10665.2
6	Mumbai	Delhi	FedEx	429	4.85	10270.6
7	Mumbai	Delhi	BlueDart	390	4.89	10401.06

8	Mumbai	Delhi	UPS	423	4.89	10330.51
9	Pune	Hyderabad	FedEx	395	4.72	10235.99
10	Pune	Hyderabad	UPS	423	4.84	10296.94
11	Pune	Hyderabad	BlueDart	412	4.93	10400.08
12	Pune	Hyderabad	DHL	449	4.95	10177.6

Findings

Chennai → Bangalore

BlueDart and UPS recorded the fastest average delivery time at **4.87 days**.

However, UPS achieved the same average delivery time at a lower average freight cost than BlueDart.

FedEx recorded the lowest average freight cost at **10,073.53**, but had a slightly higher average delivery time of **5.07 days**.

Assessment:

UPS appears to provide a strong cost-service balance for this lane, while FedEx may be suitable for cost-sensitive shipments where the small increase in average delivery time is acceptable.

Mumbai → Delhi

DHL recorded the fastest average delivery time at **4.83 days**.

FedEx followed closely at **4.85 days**, while maintaining a lower average freight cost of **10,270.60**, compared with DHL's **10,665.20**.

Assessment:

FedEx appears to offer a favourable cost-service balance on this lane because the difference in average delivery time compared with DHL is minimal while the average freight cost is lower.

Pune → Hyderabad

FedEx recorded the fastest average delivery time at **4.72 days** with an average freight cost of **10,235.99**.

DHL recorded the lowest average freight cost at **10,177.60**, but had the highest average delivery time at **4.95 days**.

The average freight-cost difference between FedEx and DHL was relatively small compared with the difference in delivery performance.

Assessment:

Based on the available measures, FedEx appears to provide a competitive cost-service balance for this transportation lane.

Overall Finding

The initial network-level carrier analysis showed very similar average delivery performance across carriers.

However, analysing performance at the **Origin–Destination–Carrier level** revealed route-specific differences in freight cost and delivery speed.

This demonstrates that carrier performance should not be evaluated solely using network-wide averages. Carrier selection may be more effective when based on transportation-lane performance and shipment service requirements.

Business Impact

A network-wide carrier strategy may overlook route-specific differences in carrier performance.

By evaluating carriers at the transportation-lane level, the organization can make more informed carrier allocation decisions based on the required balance between:

- Delivery speed
- Freight cost
- Shipment urgency
- Route-specific performance

This approach can potentially improve transportation cost efficiency while maintaining required service levels.

Recommendations

Immediate Actions

- Evaluate carrier performance at the transportation-lane level rather than relying solely on network-wide averages.
- Use faster carriers for urgent or service-critical shipments where the cost premium is justified.
- Consider lower-cost carriers for routine shipments where slightly longer delivery times are operationally acceptable.
- Review carrier allocation strategies for high-volume transportation lanes.

Medium-Term Actions

- Develop lane-level carrier performance scorecards.
- Monitor average freight cost and delivery time by carrier and route.

- Introduce additional carrier KPIs where data becomes available, such as On-Time Delivery %, damage rate, shipment reliability, and cost per weight or distance unit.
- Use Power BI to monitor cost-service trade-offs dynamically.

Data Limitations

The analysis compares carriers using the available transportation data, primarily:

- Average Delivery Days
- Average Freight Cost
- Shipment Volume

The dataset does not contain:

- Promised delivery dates
- Actual delivery dates
- Shipment weight or volume
- Distance-normalized freight cost
- Damage or claims data
- On-Time Delivery percentage

Therefore, the analysis should not be interpreted as a complete carrier performance scorecard.

Key Supply Chain Concepts Learned

- Transportation Performance Analysis
- Carrier Performance Management
- Cost-Service Trade-off
- Transportation Lane Analysis
- Carrier Allocation
- Freight Cost Analysis
- Shipment Volume Analysis
- Network-Level vs Lane-Level Analysis
- Analytical Grain
- Aggregation using AVG(), SUM(), and COUNT()
- SQL GROUP BY

Interview Talking Point

Used SQL to analyse transportation carrier performance based on shipment volume, average delivery time, and freight cost. The initial network-level analysis showed similar delivery performance across carriers, so I drilled down to the Origin–Destination–Carrier level. The lane-level analysis revealed route-specific cost-service trade-offs that were hidden by network averages, supporting a more targeted approach to carrier allocation based on shipment urgency, freight cost, and route-specific performance.

Project Architecture

CSV Files

↓

Excel Cleaning

↓

SQLite

↓

SQL Analysis

↓

Power BI

↓

Interactive Dashboards

PHASE 3: POWER BI DASHBOARD

Outcome

Developed seven interactive dashboards covering:

• Executive Overview • Inventory • Transportation • Supplier • Forecast • Order Fulfillment • Warehouse

Implemented

• 30+ reusable DAX measures • SQL validated KPIs • Star Schema • Interactive slicers • Drill-down analysis • Cross filtering • Conditional formatting • Business insights

---

FINAL DELIVERABLE

End-to-End Supply Chain Control Tower:

Tools

Excel SQL Power BI

Dataset Size

25,000 Orders

8 Business Tables

7 Interactive Dashboards

30+ DAX Measures

40+ SQL Validation Queries

Business Domains

Demand Planning

Inventory Management

Transportation Supplier Analytics

Forecasting

Warehouse Operations

Order Fulfillment

DATA MODELING:

STAR SCHEMA

Fact Tables:

Orders Forecast Transportation

Dimension Tables:

Product Supplier Warehouse Bridge Table Supplier_Product_Map

Page 1 – Executive Overview

---

1. Dashboard Objective
The Executive Overview dashboard provides senior management with a consolidated view of overall supply chain performance. It combines order fulfillment, inventory health, transportation, and supplier metrics to monitor operational efficiency and identify high-priority areas requiring business attention.

---

2. Business Questions Answered
This dashboard answers the following questions:

- How many customer orders have been processed?
- What percentage of customer demand is fulfilled?
- How many orders are delayed?
- How many inventory items require replenishment?
- What is the organization's transportation spend?

- How many suppliers support the supply chain?
- What is the average delivery lead time?
- How many SKUs are currently at inventory risk?
- How is transportation spend distributed across carriers?
- What is the overall distribution of order statuses?

---

3. KPIs

KPI	Business Meaning	DAX Measure	SQL Validation
Total Orders	Total unique customer orders processed	DISTINCTCOUNT(Orders[Order_ID])	COUNT(DISTINCT Order_ID)
Fill Rate %	Percentage of customer demand fulfilled	Delivered Qty / Demand Qty	SUM(Delivered_Qty)/SUM(Demand_Qty)
Delayed Orders	Number of delayed customer orders	COUNTROWS(DELAYED)	COUNT(*) WHERE Order_Status='DELAYED'
At Risk SKUs	Number of inventory items below safety stock	DISTINCTCOUNT(SKU_ID)	COUNT(DISTINCT SKU_ID)
Total Freight Cost	Total transportation spend	SUM(Freight_Cost)	SUM(Freight_Cost)
Total Suppliers	Active supplier base	DISTINCTCOUNT(Supplier_ID)	COUNT(DISTINCT Supplier_ID)
Average Delivery Days	Average shipment delivery time	AVERAGE(Delivery_Days)	AVG(Delivery_Days)

---

4. DAX Measures Used

1. Total Orders

```
Total Orders =
DISTINCTCOUNT(Orders[Order_ID])
```

---

2. Fill Rate %

```
Fill Rate % =
VAR Delivered = [Total Delivered]
VAR Demand = [Total Demand Qty]

RETURN
DIVIDE(Delivered, Demand, 0)
```

---

3. Delayed Orders

```
Delayed Orders =
```

```
CALCULATE(
    COUNTROWS(Orders),
    REMOVEFILTERS(Orders[Order_Status]),
    Orders[Order_Status]="DELAYED"
)
```

---

4. At Risk SKUs

```
At Risk SKUs =
CALCULATE(
    DISTINCTCOUNT(Inventory[SKU_ID]),
    Inventory[Inventory Status]="At Risk"
)
```

---

5. Total Freight Cost

```
Total Freight Cost =
SUM(Transportation[Freight_Cost])
```

---

6. Total Suppliers

```
Total Suppliers =
DISTINCTCOUNT(Supplier[Supplier_ID])
```

---

7. Average Delivery Days

```
Avg Delivery Days =
AVERAGE(Transportation[Delivery_Days])
```

---

5. SQL Validation Queries

KPI 1 – Total Orders

```
SELECT COUNT(DISTINCT Order_ID) AS Total_Orders
FROM Orders;
```

KPI 2 – Fill Rate %

```
SELECT
ROUND(
SUM(Delivered_Qty)*100.0/
SUM(Demand_Qty),2
) AS Fill_Rate
FROM Orders;
```

KPI 3 – Delayed Orders

```
SELECT
COUNT(*) AS Delayed_Orders
FROM Orders
WHERE Order_Status='DELAYED';
```

KPI 4 – At Risk SKUs

```
SELECT
COUNT(DISTINCT SKU_ID) AS At_Risk_SKUs
FROM Inventory
```

```
WHERE Current_Stock < Safety_Stock;
```

KPI 5 – Total Freight Cost

```
SELECT
SUM(Freight_Cost) AS Total_Freight_Cost
FROM Transportation;
```

KPI 6 – Total Suppliers

```
SELECT
COUNT(DISTINCT Supplier_ID) AS Total_Suppliers
FROM Supplier;
```

KPI 7 – Average Delivery Days

```
SELECT
ROUND(AVG(Delivery_Days),2) AS Avg_Delivery_Days
FROM Transportation;
```

---

6. Visual Documentation

Visual 1 – Order Status Distribution

Objective

Shows the proportion of Delivered, Partial, and Delayed orders.

Business Insight

Helps management understand the overall order fulfillment performance.

DAX Used

Total Orders

SQL

```
SELECT
Order_Status,
COUNT(DISTINCT Order_ID) AS Orders
FROM Orders
GROUP BY Order_Status;
```

Visual 2 – Inventory Risk by Warehouse

Objective

Shows warehouses with the highest number of inventory items below safety stock.

Business Insight

Helps prioritize replenishment activities.

DAX Used

At Risk SKUs

SQL

```
SELECT
Warehouse_ID,
COUNT(DISTINCT SKU_ID) AS At_Risk_SKUs
FROM Inventory
WHERE Current_Stock < Safety_Stock
GROUP BY Warehouse_ID
ORDER BY At_Risk_SKUs DESC;
```

Visual 3 – Freight Cost by Carrier

Objective

Compares logistics expenditure across transportation carriers.

Business Insight

Identifies carriers contributing the highest logistics cost.

DAX Used

Total Freight Cost

SQL

```
SELECT
Carrier,
SUM(Freight_Cost) AS Freight_Cost
FROM Transportation
GROUP BY Carrier
ORDER BY Freight_Cost DESC;
```

---

7. Executive Insights

- **25K** customer orders were processed across the supply chain.
- Overall **Fill Rate** is **66.9%**, indicating opportunities to improve order fulfillment.
- **8K** customer orders experienced delivery delays.
- **234 SKUs** are currently below safety stock and require replenishment.
- Total transportation expenditure reached **₹52 Million**.
- The supply network is supported by **50 suppliers**.
- Average transportation lead time is **4.89 days**.
- **WH7** has the highest inventory risk and should be prioritized for inventory replenishment.
- **DHL** accounts for the highest freight expenditure, presenting an opportunity for carrier cost optimization.

---

8. Key Business Findings

Finding	Business Impact
Fill Rate below 70%	Customer service improvement opportunity
WH7 has highest inventory risk	Immediate replenishment required
DHL contributes highest freight cost	Transportation optimization opportunity
8K delayed orders	Service level improvement required
50 active suppliers	Strong supplier network
Average delivery time under 5 days	Transportation performance is relatively stable

---

9. Skills Demonstrated

- Power BI Dashboard Design
- KPI Development
- DAX Measures
- SQL Validation
- Supply Chain Analytics
- Inventory Analysis
- Transportation Cost Analysis

- Executive Reporting
- Data Modeling
- Business Intelligence

Page 2 – Inventory Dashboard

---

1. Dashboard Objective

The Inventory Dashboard provides a comprehensive view of inventory health across warehouses by monitoring stock availability, inventory investment, replenishment risk, and supplier lead times. It helps warehouse managers and planners identify stock shortages, optimize inventory investment, and prioritize replenishment activities.

---

2. Business Questions Answered

This dashboard answers the following questions:

- How much inventory is currently available?
- How much safety stock is maintained?
- What is the total inventory value?
- How many SKUs are below safety stock?
- How many SKUs are healthy?
- What is the average supplier lead time?
- Which warehouses have the highest inventory?
- Which warehouses carry the highest inventory investment?
- What percentage of inventory is healthy versus at risk?

---

3. KPIs

KPI	Business Meaning	DAX Measure	SQL Validation
Total Current Stock	Total inventory currently available	SUM(Current_Stock)	SUM(Current_Stock)
Total Safety Stock	Total minimum stock maintained	SUM(Safety_Stock)	SUM(Safety_Stock)
Total Inventory Value	Monetary value of inventory	SUM(Inventory_Value)	SUM(Inventory_Value)
At Risk SKUs	Distinct SKUs below safety stock	DISTINCTCOUNT(SKU_ID)	COUNT(DISTINCT SKU_ID)
Healthy SKUs	Inventory records meeting safety stock requirement	COUNTROWS()	COUNT(*)
Average Lead Time	Average supplier replenishment lead time	AVERAGE(Lead_Time_Days)	AVG(Lead_Time_Days)

---

4. DAX Measures Used

KPI 1 – Total Current Stock

```
Total Current Stock =
SUM(Inventory[Current_Stock])
```

KPI 2 – Total Safety Stock

```
Total Safety Stock =
SUM(Inventory[Safety_Stock])
```

KPI 3 – Total Inventory Value

```
Total Inventory Value =
SUM(Inventory[Inventory_Value])
```

KPI 4 – At Risk SKUs

```
At Risk SKUs =
CALCULATE(
    DISTINCTCOUNT(Inventory[SKU_ID]),
    Inventory[Inventory Status] = "At Risk"
)
```

KPI 5 – Healthy SKUs

```
Healthy SKUs =
CALCULATE(
    DISTINCTCOUNT(Inventory[SKU_ID]),
    Inventory[Inventory Status] = "Healthy"
)
```

KPI 6 – Average Lead Time

```
Avg Lead Time =
AVERAGE(Supplier[Lead_Time_Days])
```

---

5. SQL Validation Queries

KPI 1 – Total Current Stock

```
SELECT
SUM(Current_Stock) AS Total_Current_Stock
FROM Inventory;
```

KPI 2 – Total Safety Stock

```
SELECT
SUM(Safety_Stock) AS Total_Safety_Stock
FROM Inventory;
```

KPI 3 – Total Inventory Value

```
SELECT
SUM(Inventory_Value) AS Total_Inventory_Value
FROM Inventory;
```

KPI 4 – At Risk SKUs

```
SELECT
COUNT(DISTINCT SKU_ID) AS At_Risk_SKUs
FROM Inventory
WHERE Current_Stock < Safety_Stock;
```

KPI 5 – Healthy SKUs

```
SELECT
COUNT(*) AS Healthy_SKUs
```

```sql
FROM Inventory
WHERE Current_Stock >= Safety_Stock;
```

KPI 6 – Average Lead Time

```sql
SELECT
ROUND(AVG(Lead_Time_Days),2) AS Avg_Lead_Time
FROM Supplier;
```

---

6. Visual Documentation

Visual 1 – Inventory Status Distribution

Objective

Displays the proportion of Healthy and At Risk inventory across all warehouses.

Business Insight

Provides an overall inventory health indicator, helping management quickly assess replenishment risk.

DAX Used

```
Inventory Status =
IF(
    Inventory[Current_Stock] < Inventory[Safety_Stock],
    "At Risk",
    "Healthy"
)
```

SQL Validation

```sql
SELECT
CASE
    WHEN Current_Stock < Safety_Stock THEN 'At Risk'
    ELSE 'Healthy'
END AS Inventory_Status,
COUNT(*) AS SKU_Count
FROM Inventory
GROUP BY Inventory_Status;
```

Visual 2 – Stock Availability by Warehouse

Objective

Compares Current Stock with Safety Stock for each warehouse.

Business Insight

Highlights whether warehouses maintain sufficient stock above minimum inventory requirements.

DAX Used

Total Current Stock

Total Safety Stock

SQL Validation

```sql
SELECT
Warehouse_ID,
SUM(Current_Stock) AS Current_Stock,
SUM(Safety_Stock) AS Safety_Stock
FROM Inventory
GROUP BY Warehouse_ID
```

```sql
ORDER BY Current_Stock DESC;
```

Visual 3 – Inventory Value by Warehouse

Objective

Shows the total inventory investment by warehouse.

Business Insight

Identifies warehouses with the highest capital tied up in inventory.

DAX Used

Total Inventory Value

SQL Validation

```sql
SELECT
Warehouse_ID,
SUM(Inventory_Value) AS Inventory_Value
FROM Inventory
GROUP BY Warehouse_ID
ORDER BY Inventory_Value DESC;
```

---

7. Executive Insights

- Total current stock across all warehouses is approximately **2M units.**
- Safety stock maintained across the network totals **523K units.**
- Total inventory investment is approximately **₹463M.**
- **234 SKUs** are currently below safety stock and require replenishment.
- Approximately **2.68K SKUs (89.17%)** are operating within healthy inventory levels.
- The average supplier lead time across all SKUs is **14.66 days**, supporting replenishment planning.
- **WH3** and **WH5** hold the highest inventory investment (approximately **₹61M** each).
- Inventory health across warehouses remains strong, with nearly **90%** of inventory above safety stock.

---

8. Key Business Findings

Finding	Business Impact
Inventory Health = 89%	Inventory is generally well maintained
234 At Risk SKUs	Immediate replenishment required
₹463M inventory investment	High working capital commitment
WH3 & WH5 hold highest inventory value	Monitor inventory carrying costs
Average Lead Time = 14.7 days	Supports replenishment planning
Current stock significantly exceeds safety stock	Indicates stable inventory availability

---

9. Skills Demonstrated

- Inventory Analytics
- Warehouse Performance Analysis
- Inventory Health Monitoring
- DAX Measures
- SQL Aggregation & Validation
- Inventory Valuation
- Safety Stock Analysis

- Supply Chain KPI Development
- Executive Dashboard Design
- Business Intelligence Reporting

Page 3 – Transportation Dashboard

Dashboard Objective

Monitor transportation performance by analyzing shipment volume, freight expenditure, carrier utilization, delivery performance, and transportation lanes to support logistics cost optimization and carrier performance management.

Business Questions Answered

- How many shipments were handled?
- How much was spent on freight?
- Which carrier incurs the highest transportation cost?
- Which carrier handles the most shipments?
- What is the average transportation lead time?
- Which transportation lanes perform best?
- How many carriers and routes are currently utilized?

KPI Cards

---

1. Total Shipments

Business Purpose

Displays the total number of shipments handled.

DAX

```
Total Shipments =
COUNTROWS(Transportation)
```

SQL Validation

```
SELECT COUNT(*) AS Total_Shipments
FROM Transportation;
```

---

2. Total Freight Cost

Business Purpose

Displays total transportation expenditure.

DAX

```
Total Freight Cost =
SUM(Transportation[Freight_Cost])
```

SQL Validation

```
SELECT
SUM(Freight_Cost) AS Total_Freight_Cost
FROM Transportation;
```

---

3. Average Freight Cost

Business Purpose

Measures the average transportation cost per shipment.

DAX

```
Avg Freight Cost =
AVERAGE(Transportation[Freight_Cost])
```

SQL Validation

```
SELECT
AVG(Freight_Cost) AS Avg_Freight_Cost
FROM Transportation;
```

---

4. Average Delivery Days

Business Purpose

Measures average transportation lead time.

DAX

```
Avg Delivery Days =
AVERAGE(Transportation[Delivery_Days])
```

SQL Validation

```
SELECT
AVG(Delivery_Days) AS Avg_Delivery_Days
FROM Transportation;
```

---

5. Total Carriers

Business Purpose

Displays the number of logistics partners.

DAX

```
Total Carriers =
DISTINCTCOUNT(Transportation[Carrier])
```

SQL Validation

```
SELECT
COUNT(DISTINCT Carrier) AS Total_Carriers
FROM Transportation;
```

---

6. Total Routes

Business Purpose

Shows the number of transportation lanes available.

DAX

```
Total Routes =
DISTINCTCOUNT(Transportation[Origin] & " → " & Transportation[Destination])
```

SQL Validation

```
SELECT COUNT(DISTINCT Origin || ' → ' || Destination) AS Total_Routes FROM Transportation;
```

Visual 1 – Freight Spend by Carrier

Objective

Compare transportation spending across logistics partners.

Business Purpose

Identify the most expensive carrier and evaluate freight cost distribution.

DAX Used

```
Total Freight Cost =
SUM(Transportation[Freight_Cost])
```

SQL Validation

```
SELECT
Carrier,
SUM(Freight_Cost) AS Total_Freight_Cost
FROM Transportation
```

```
GROUP BY Carrier
ORDER BY Total_Freight_Cost DESC;
```

Visual 2 – Shipment Count by Carrier

Objective

Compare shipment volume handled by each carrier.

Business Purpose

Evaluate carrier utilization and identify dependency on specific logistics partners.

DAX Used

```
Shipment Count =
COUNTROWS(Transportation)
(or)
Shipment Count =
DISTINCTCOUNT(Transportation[Shipment_ID])
```

SQL Validation

```
SELECT
Carrier,
COUNT(*) AS Shipment_Count
FROM Transportation
GROUP BY Carrier
ORDER BY Shipment_Count DESC;
```

Visual 3 – Carrier Performance by Lane (Matrix)

Objective

Evaluate carrier performance across transportation lanes.

Business Purpose

Compare average delivery time by Origin (or Route) and Carrier to identify the best-performing carrier for each lane.

DAX Used

```
Avg Delivery Days =
AVERAGE(Transportation[Delivery_Days])
```

SQL Validation

```
SELECT
Origin,
Carrier,
ROUND(AVG(Delivery_Days),2) AS Avg_Delivery_Days
FROM Transportation
GROUP BY Origin, Carrier
ORDER BY Origin, Carrier;
```

Skills Demonstrated

Power BI

- KPI Card Design
- Clustered Bar Chart
- Horizontal Bar Chart
- Matrix Visualization
- Interactive Slicers
- Cross-filtering

- Dashboard Design
- Data Modeling

DAX

- COUNTROWS()
- DISTINCTCOUNT()
- SUM()
- AVERAGE()
- Measure Creation
- Aggregation Functions

SQL

- COUNT()
- SUM()
- AVG()
- GROUP BY
- DISTINCT
- ORDER BY

Supply Chain Concepts

- Transportation Analytics
- Freight Cost Analysis
- Carrier Performance Evaluation
- Shipment Volume Analysis
- Route Performance Monitoring
- Logistics Cost Optimization
- Carrier Utilization Analysis

Key Business Findings

KPI Findings

- Approximately **5,000 shipments** were executed across the logistics network.
- Total freight expenditure reached **₹52M.**
- Average freight cost per shipment was **₹10.41K.**
- Average delivery performance remained stable at **4.89 days.**
- Transportation operations utilized **4 logistics carriers** across **3 transportation routes**.

Freight Spend by Carrier

- **DHL** recorded the highest freight expenditure (~₹13.2M).
- Freight costs are relatively balanced across carriers, indicating diversified transportation spending.
- Carrier cost optimization opportunities exist through contract review and shipment allocation.

Shipment Count by Carrier

- Shipment volume distribution highlights carrier utilization across the network.
- Carrier workload appears relatively balanced, reducing dependency on a single logistics provider.
- Shipment allocation analysis supports better carrier capacity planning.

Carrier Performance by Lane

- Delivery performance differs across transportation lanes.

- Certain carrier–route combinations consistently achieve lower delivery times.
- Lane-specific carrier assignment can improve service levels and transportation efficiency.

Business Decisions Supported
This dashboard enables transportation managers to:

- Monitor transportation expenditure.
- Evaluate carrier utilization.
- Compare carrier delivery performance.
- Optimize shipment allocation across carriers.
- Reduce freight costs.
- Improve transportation service levels.
- Support carrier contract negotiations.
- Optimize transportation network planning.

Executive Summary
The Transportation Dashboard provides a comprehensive view of logistics operations by integrating shipment volume, freight spending, carrier performance, and route efficiency into a single decision-support interface. By combining operational KPIs with carrier-level analytics and SQL-validated Power BI measures, the dashboard enables logistics teams to monitor transportation costs, evaluate carrier effectiveness, identify optimization opportunities, and make data-driven decisions that improve service performance while controlling logistics expenditure.

Page 4 – Supplier Dashboard
Dashboard Objective
Evaluate supplier performance by analyzing supplier reliability, customer demand fulfillment, on-time delivery (OTD), lead time, defect rate, and supplier risk to support supplier selection, performance improvement, and procurement decisions.

Business Questions Answered

- How many suppliers support the supply chain?
- Which suppliers fulfill the highest customer demand?
- Which suppliers have the highest On-Time Delivery (OTD)?
- What is the average supplier lead time?
- What is the average supplier defect rate?
- Which suppliers are classified as high risk?
- Which suppliers require immediate performance improvement?

KPI Cards

---

1. Total Suppliers
Business Purpose
Displays the total number of active suppliers.
DAX

```
Total Suppliers =
DISTINCTCOUNT(Supplier[Supplier_ID])
```

SQL Validation

```sql
SELECT
COUNT(DISTINCT Supplier_ID) AS Total_Suppliers
FROM Supplier;
```

---

2. Total Demand Qty

Business Purpose

Displays the total customer demand supported by suppliers.

DAX

```
Total Demand Qty =
SUM(Orders[Demand_Qty])
```

SQL Validation

```sql
SELECT
SUM(Demand_Qty) AS Total_Demand_Qty
FROM Orders;
```

---

3. Average OTD %

Business Purpose

Measures supplier delivery reliability.

DAX

```
Average OTD % =
DIVIDE(
    AVERAGE(Supplier[OTD_Pct]),
    100
)
```

SQL Validation

```sql
SELECT
AVG(OTD_Pct) AS Avg_OTD_Percentage
FROM Supplier;
```

---

4. Average Lead Time

Business Purpose

Measures the average supplier replenishment lead time.

DAX

```
Avg Lead Time =
AVERAGE(Supplier[Lead_Time_Days])
```

SQL Validation

```sql
SELECT
AVG(Lead_Time_Days) AS Avg_Lead_Time
FROM Supplier;
```

---

5. Average Defect Rate

Business Purpose

Measures average supplier quality performance.

DAX

```
Average Defect Rate =
DIVIDE(
    AVERAGE(Supplier[Defect_Rate]),
    100
```

)

SQL Validation

```
SELECT
AVG(Defect_Rate) AS Avg_Defect_Rate
FROM Supplier;
```

---

6. High Risk Suppliers

Business Purpose

Identifies suppliers requiring immediate performance review based on service and quality thresholds.

Risk Criteria

- OTD < 75%
- Lead Time > 20 Days
- Defect Rate > 5%

DAX

```
High Risk Suppliers =
CALCULATE(
    DISTINCTCOUNT(Supplier[Supplier_ID]),
    FILTER(
        Supplier,
        Supplier[OTD_Pct] < 75 ||
        Supplier[Lead_Time_Days] > 20 ||
        Supplier[Defect_Rate] > 5
    )
)
```

SQL Validation

```
SELECT
COUNT(DISTINCT Supplier_ID) AS High_Risk_Suppliers
FROM Supplier
WHERE
OTD_Pct < 75
OR Lead_Time_Days > 20
OR Defect_Rate > 5;
```

Visual 1 – Top 10 Suppliers by Customer Demand

Objective

Identify suppliers supporting the highest customer demand.

Business Purpose

Highlights suppliers with the greatest contribution to customer demand, helping identify strategic suppliers and potential dependency risks.

DAX Used

```
Total Demand Qty =
SUM(Orders[Demand_Qty])
```

SQL Validation

```
SELECT
    s.Supplier_Name,
    SUM(o.Demand_Qty) AS Total_Demand
FROM Orders o
```

```
JOIN Supplier_Product_Map sp
    ON o.SKU_ID = sp.SKU_ID
JOIN Supplier s
    ON sp.Supplier_ID = s.Supplier_ID
GROUP BY s.Supplier_Name
ORDER BY Total_Demand DESC
LIMIT 10;
```

Visual 2 – Top 10 Suppliers by OTD %

Objective

Compare supplier delivery reliability.

Business Purpose

Identifies the most reliable suppliers based on On-Time Delivery performance.

DAX Used

```
Average OTD % =
DIVIDE(
AVERAGE(Supplier[OTD_Pct]),
100
)
```

SQL Validation

```
SELECT
Supplier_Name,
OTD_Pct
FROM Supplier
ORDER BY OTD_Pct DESC
LIMIT 10;
```

Visual 3 – Supplier Risk Matrix

Objective

Evaluate supplier risk using Lead Time and Defect Rate.

Business Purpose

Identify suppliers with both long lead times and high defect rates for proactive supplier development and risk mitigation.

DAX Used

None

(Scatter chart uses raw columns.)

SQL Validation

```
SELECT
Supplier_Name,
Lead_Time_Days,
Defect_Rate
FROM Supplier;
```

Skills Demonstrated

Power BI

- KPI Card Design
- Clustered Column Charts
- Scatter Plot (Risk Matrix)

- Top N Filtering
- Interactive Slicers
- Cross-filtering
- Dashboard Layout Design
- Conditional Formatting

DAX

- DISTINCTCOUNT()
- SUM()
- AVERAGE()
- DIVIDE()
- CALCULATE()
- FILTER()
- Logical Conditions (OR)

SQL

- COUNT(DISTINCT)
- SUM()
- AVG()
- GROUP BY
- ORDER BY
- LIMIT / TOP
- WHERE Filtering
- JOIN Operations

Supply Chain Concepts

- Supplier Performance Management
- Supplier Risk Assessment
- Supplier Quality Analysis
- On-Time Delivery (OTD)
- Procurement Analytics
- Vendor Performance Monitoring
- Supplier Segmentation
- Strategic Sourcing

Key Business Findings

KPI Findings

- The supply base consists of **50 suppliers** supporting approximately **4M units** of customer demand.
- Suppliers achieved an average **OTD of 77.98%**, indicating opportunities to improve delivery reliability.
- Average supplier lead time is **14.7 days**.
- Average supplier defect rate stands at **4.61%**.
- **38 suppliers** meet at least one high-risk criterion, indicating significant supplier performance improvement opportunities.

Top Suppliers by Customer Demand

- **Supplier_38** supports the highest customer demand (~144K units).
- Customer demand is concentrated among a limited number of suppliers, indicating potential dependency risk.
- High-volume suppliers should receive priority in supplier relationship management.

Top Suppliers by OTD

- The highest-performing suppliers consistently achieve **90–97% On-Time Delivery**.
- These suppliers represent benchmarks for delivery performance.
- Strong OTD suppliers should be prioritized for critical procurement requirements.

Supplier Risk Matrix

- Suppliers located in the **upper-right quadrant** (high lead time and high defect rate) represent the greatest operational risk.
- Suppliers with low lead time and low defect rates demonstrate stable and reliable performance.
- The matrix enables targeted supplier development rather than treating all suppliers equally.

Business Decisions Supported

This dashboard enables procurement and sourcing teams to:

- Evaluate supplier reliability.
- Identify high-risk suppliers.
- Reduce supplier-related disruptions.
- Improve supplier quality performance.
- Prioritize supplier development initiatives.
- Reduce procurement risk.
- Strengthen strategic sourcing decisions.
- Monitor supplier performance continuously.

Executive Summary

The Supplier Dashboard provides a comprehensive view of supplier performance by integrating customer demand, delivery reliability, lead time, product quality, and supplier risk into a single decision-support platform. SQL-validated Power BI measures enable procurement teams to identify strategic suppliers, monitor operational risk, improve supplier performance, and make data-driven sourcing decisions that enhance supply chain resilience and service reliability.

Page 5 – Forecast Dashboard

Dashboard Objective

Evaluate forecast planning performance by comparing forecasted demand with actual demand, measuring forecast accuracy, identifying forecast variance across warehouses and months, and supporting demand planning improvements.

Business Questions Answered

- How much demand was forecasted?
- How much demand actually occurred?
- What is the overall forecast accuracy?
- What is the forecast error?
- Which warehouse has the highest forecast accuracy?
- Which months experienced over-forecasting or under-forecasting?
- How many warehouses are included in the planning network?

KPI Cards

---

1. Total Forecast Qty

Business Purpose

Displays the total forecasted demand.

DAX

```
Total Forecast Qty =
SUM(Forecast[Forecast_Qty])
```

SQL Validation

```
SELECT
SUM(Forecast_Qty) AS Total_Forecast_Qty
FROM Forecast;
```

---

2. Total Actual Demand

Business Purpose

Displays total customer demand recorded.

DAX

```
Total Actual Demand =
SUM(Forecast[Actual_Demand])
```

SQL Validation

```
SELECT
SUM(Actual_Demand) AS Total_Actual_Demand
FROM Forecast;
```

---

3. Forecast Variance

Business Purpose

Measures deviation between forecast and actual demand.

Positive Value → Over Forecast

Negative Value → Under Forecast

DAX

```
Forecast Variance =
[Total Forecast Qty] -
[Total Actual Demand]
```

SQL Validation

```
SELECT
SUM(Forecast_Qty) -
SUM(Actual_Demand) AS Forecast_Variance
FROM Forecast;
```

---

4. Forecast Accuracy %

Business Purpose

Measures how closely forecast matches actual demand.

Higher percentage indicates better demand planning performance.

DAX

```
Forecast Accuracy % =

VAR ForecastError =
DIVIDE(
ABS([Forecast Variance]),
[Total Actual Demand]
```

```
)

VAR Accuracy =
1 - ForecastError

RETURN
MAX(0,Accuracy)
```

SQL Validation

```
SELECT
    ROUND(
        (
            1 -
            (
                ABS(SUM(Forecast_Qty) - SUM(Actual_Demand))
                * 100.0
                / SUM(Actual_Demand)
            ) / 100
        ) * 100,
        2
    ) AS Forecast_Accuracy_Pct
FROM Forecast;
```

---

5. Forecast Error %

Business Purpose

Shows forecasting deviation percentage.
Lower values indicate better planning accuracy.

DAX

```
Forecast Error % =
DIVIDE(
ABS([Forecast Variance]),
[Total Actual Demand]
)
```

SQL Validation

```
SELECT
ABS(
SUM(Forecast_Qty)-SUM(Actual_Demand)
)
/SUM(Actual_Demand)
AS Forecast_Error
FROM Forecast;
```

---

6. Total Warehouses

Business Purpose

Displays the number of warehouses included in forecasting.

DAX

```
Total Warehouses =
DISTINCTCOUNT(Inventory[Warehouse_ID])
```

SQL Validation

```
SELECT
COUNT(DISTINCT Warehouse_ID)
AS Total_Warehouses
FROM Inventory;
```

Visual 1 – Forecast vs Actual Demand

Objective

Compare planned demand against actual demand across warehouses.

Business Purpose

Highlights forecast deviation at warehouse level and identifies locations requiring planning improvements.

DAX Used

Total Forecast Qty

Total Actual Demand

SQL Validation

```
SELECT
Warehouse_ID,
SUM(Forecast_Qty) AS Forecast_Qty,
SUM(Actual_Demand) AS Actual_Demand
FROM Forecast
GROUP BY Warehouse_ID
ORDER BY Actual_Demand DESC;
```

Visual 2 – Forecast Accuracy by Warehouse

Objective

Compare planning accuracy across warehouses.

Business Purpose

Identifies warehouses with the most reliable demand forecasting.

DAX Used

Forecast Accuracy %

SQL Validation

```
SELECT
    Warehouse_ID,

    ROUND(
        CASE
            WHEN (
                1 -
                (
                    ABS(SUM(Forecast_Qty) - SUM(Actual_Demand))
                    * 1.0
                    / SUM(Actual_Demand)
                )
            ) < 0
            THEN 0

            ELSE (
```

```
                1 -
                (
                    ABS(SUM(Forecast_Qty) - SUM(Actual_Demand))
                    * 1.0
                    / SUM(Actual_Demand)
                )
            )
        END * 100,
        2
    ) AS Forecast_Accuracy_Pct

FROM Forecast

GROUP BY Warehouse_ID

ORDER BY Forecast_Accuracy_Pct DESC;
```

Visual 3 – Monthly Forecast Variance

Objective

Track forecasting deviation across months.

Business Purpose

Identifies months where forecasting significantly exceeded or fell below actual demand.

DAX Used

Forecast Variance

SQL Validation

```
SELECT

Month,

SUM(Forecast_Qty)-SUM(Actual_Demand)
AS Forecast_Variance

FROM Forecast

GROUP BY Month

ORDER BY
CASE Month
WHEN 'Jan' THEN 1
WHEN 'Feb' THEN 2
WHEN 'Mar' THEN 3
WHEN 'Apr' THEN 4
WHEN 'May' THEN 5
WHEN 'Jun' THEN 6
END;
```

Skills Demonstrated

Power BI

- KPI Card Design
- Clustered Column Charts
- Horizontal Bar Charts
- Variance Analysis
- Conditional Formatting
- Interactive Slicers
- Dashboard Design
- Cross-filtering

DAX

- SUM()
- DISTINCTCOUNT()
- DIVIDE()
- ABS()
- VAR
- MAX()
- Measure Creation
- Forecast Accuracy Calculation

SQL

- SUM()
- COUNT(DISTINCT)
- GROUP BY
- ORDER BY
- CASE
- Mathematical Expressions
- Aggregate Functions

Supply Chain Concepts

- Demand Planning
- Forecast Accuracy
- Forecast Error Analysis
- Forecast Variance
- Warehouse Performance
- Sales & Operations Planning (S&OP)
- Demand Analytics
- Inventory Planning

Key Business Findings

KPI Findings

- Total forecast volume closely matched actual demand at approximately **1M units,** resulting in a **forecast variance of only -1K units.**
- Overall **forecast accuracy reached 99.92%,** indicating highly reliable demand planning.
- Forecast error remained exceptionally low at **0.08%.**
- Forecast planning covered **8 warehouses** across the distribution network.

Forecast vs Actual Demand

- Forecasted demand closely aligned with actual demand across all warehouses.
- Minor deviations suggest effective planning and stable customer demand.
- No warehouse experienced significant over-forecasting or under-forecasting.

Forecast Accuracy by Warehouse

- **WH8 achieved the highest forecast accuracy (99.92%)**, demonstrating excellent planning performance.
- All warehouses maintained forecast accuracy above **95%**, indicating consistent planning quality across the network.

Monthly Forecast Variance

- **April recorded the highest positive forecast variance (+9.6K)**, indicating over-forecasting.
- **May experienced the largest negative variance (-8.4K)**, indicating under-forecasting.
- Monthly variance remained relatively small overall, reflecting stable forecasting performance.

Business Decisions Supported

This dashboard enables planning teams to:

- Monitor forecast reliability.
- Improve demand planning accuracy.
- Identify warehouse-level forecasting issues.
- Detect seasonal planning deviations.
- Reduce forecast error.
- Improve inventory planning.
- Support Sales & Operations Planning (S&OP).
- Optimize replenishment planning.

Executive Summary

The Forecast Dashboard provides a comprehensive assessment of demand planning performance by comparing forecasted and actual demand across warehouses and time periods. Through SQL-validated Power BI measures, it evaluates forecast accuracy, variance, and planning error to identify deviations that impact inventory and replenishment decisions. Warehouse-level accuracy comparisons and monthly variance trends enable planners to continuously refine forecasting models, reduce planning error, and support more effective Sales & Operations Planning (S&OP), resulting in improved inventory availability and operational efficiency

Page 6 – Order Fulfillment Dashboard

Dashboard Objective

Monitor order fulfillment performance by evaluating customer demand, delivered quantity, fill rate, delayed orders, regional performance, warehouse service levels, and product-level fulfillment to improve customer service and inventory availability.

Business Questions Answered

- How many customer orders were processed?
- What is the total customer demand?
- How much demand was fulfilled?
- What is the overall fill rate?
- What percentage of orders were delayed?
- Which regions require fulfillment improvement?
- Which warehouses perform best?
- Which product categories and products have the lowest fill rates?

KPI Cards

---

1. Total Orders

Business Purpose

Displays the total number of customer orders processed.

DAX

```
Total Orders =
DISTINCTCOUNT(Orders[Order_ID])
```

SQL Validation

```
SELECT
COUNT(DISTINCT Order_ID) AS Total_Orders
FROM Orders;
```

---

2. Total Demand Qty

Business Purpose

Displays total customer demand.

DAX

```
Total Demand Qty =
SUM(Orders[Demand_Qty])
```

SQL Validation

```
SELECT
SUM(Demand_Qty) AS Total_Demand_Qty
FROM Orders;
```

---

3. Total Delivered

Business Purpose

Displays total quantity successfully delivered.

DAX

```
Total Delivered =
SUM(Orders[Delivered_Qty])
```

SQL Validation

```
SELECT
SUM(Delivered_Qty) AS Total_Delivered
FROM Orders;
```

---

4. Fill Rate %

Business Purpose

Measures the percentage of customer demand successfully fulfilled.
Higher values indicate better order fulfillment performance.

DAX

```
Fill Rate % =
VAR Delivered = [Total Delivered]
VAR Demand = [Total Demand Qty]

RETURN
DIVIDE(Delivered, Demand, 0)
```

SQL Validation

```
SELECT
    ROUND(
        SUM(Delivered_Qty) * 100.0 / SUM(Demand_Qty),
        2
    ) AS Fill_Rate_Pct
FROM Orders;
```

---

5. Delayed Order %

Business Purpose

Measures the percentage of delayed customer orders.
Lower values indicate better service performance.

DAX

```
Delayed Order % =
DIVIDE(
    [Delayed Orders],
    CALCULATE(
        [Total Orders],
        REMOVEFILTERS(Orders[Order_Status])
    ),
    0
)
```

SQL Validation

```
SELECT
COUNT(CASE WHEN Order_Status='DELAYED' THEN 1 END)*100.0/
COUNT(*) AS Delayed_Order_Percentage
FROM Orders;
```

---

6. Regions Served

Business Purpose

Displays the number of customer regions served.

DAX

```
Regions Served =
DISTINCTCOUNT(Orders[Customer_Region])
```

SQL Validation

```
SELECT
COUNT(DISTINCT Customer_Region) AS Regions_Served
FROM Orders;
```

Visual 1 – Regional Demand vs Delivered Quantity

Objective

Compare customer demand and delivered quantity across regions.

Business Purpose

Highlights fulfillment gaps between customer demand and delivered quantity.

DAX Used

Total Demand Qty

Total Delivered

SQL Validation

```
SELECT
Customer_Region,
SUM(Demand_Qty) AS Total_Demand,
SUM(Delivered_Qty) AS Total_Delivered
FROM Orders
GROUP BY Customer_Region
ORDER BY Total_Demand DESC;
```

Visual 2 – Fill Rate % by Product Category

Objective

Compare fulfillment performance across product categories.

Business Purpose

Identifies categories with lower service levels that require inventory or replenishment improvements.

DAX Used

Fill Rate %

SQL Validation

```
SELECT
    P.Category,
    ROUND(
        SUM(O.Delivered_Qty) * 100.0 /
        SUM(O.Demand_Qty),
        2
    ) AS Fill_Rate
FROM Orders O
JOIN Product P
    ON O.SKU_ID = P.SKU_ID
GROUP BY P.Category
ORDER BY Fill_Rate DESC;
```

Visual 3 – Warehouse Fill Rate %

Objective

Compare warehouse fulfillment performance.

Business Purpose

Identifies warehouses delivering the highest customer service levels.

DAX Used

Fill Rate %

SQL Validation

```
SELECT
    Warehouse_ID,
    ROUND(
        SUM(Delivered_Qty) * 100.0 /
        SUM(Demand_Qty),
        2
    ) AS Fill_Rate_Pct
FROM Orders
GROUP BY Warehouse_ID
```

```
ORDER BY Fill_Rate_Pct DESC;
```

Visual 4 – Lowest Fill Rate Products

Objective

Identify products with the poorest fulfillment performance.

Business Purpose

Highlights products requiring inventory optimization, supplier review, or replenishment planning.

DAX Used

Fill Rate %

SQL Validation

```
SELECT
    P.Product_Name,
    ROUND(
        SUM(O.Delivered_Qty) * 100.0 /
        SUM(O.Demand_Qty),
        2
    ) AS Fill_Rate_Pct
FROM Orders O
JOIN Product P
    ON O.SKU_ID = P.SKU_ID
GROUP BY P.Product_Name
ORDER BY Fill_Rate_Pct ASC
LIMIT 10;
```

Skills Demonstrated

Power BI

- KPI Card Design
- Clustered Column Chart
- Horizontal Bar Charts
- Ranking (Top/Bottom N)
- Interactive Slicers
- Cross-filtering
- Dashboard Layout Design
- Conditional Formatting

DAX

- DISTINCTCOUNT()
- SUM()
- DIVIDE()
- CALCULATE()
- REMOVEFILTERS()
- VAR
- Measure Creation

SQL

- COUNT(DISTINCT)
- SUM()
- GROUP BY
- ORDER BY

- LIMIT / TOP
- CASE
- Aggregate Functions

Supply Chain Concepts

- Order Fulfillment
- Fill Rate Analysis
- Customer Service Level
- Warehouse Performance
- Regional Demand Analysis
- Product Availability
- Delivery Performance
- Inventory Optimization

Key Business Findings

KPI Findings

- The business processed approximately **25K customer orders**, fulfilling **4M units of demand.**
- Around **3M units** were successfully delivered, resulting in an overall **Fill Rate of 66.9%.**
- Approximately **33.12%** of orders experienced delivery delays.
- Operations supported customers across **4 geographic regions.**

Regional Demand vs Delivered Quantity

- All regions exhibit a noticeable gap between demand and delivered quantity.
- **North** recorded the highest demand volume, followed closely by **West** and **East**.
- The consistent gap across regions suggests network-wide fulfillment constraints rather than a region-specific issue.

Fill Rate by Product Category

- Fill rate performance remains relatively consistent across product categories (**66–68%**).
- **Furniture** achieved the highest fill rate (~67.5%), while **Appliances** recorded the lowest.
- No category shows extreme underperformance, indicating balanced inventory allocation.

Warehouse Fill Rate

- Warehouse service levels range between **66.2% and 68.0%.**
- **WH5** achieved the highest fulfillment performance, while **WH2** recorded the lowest.
- Warehouse performance is generally stable, though lower-performing sites present opportunities for process improvement.

Lowest Fill Rate Products

- Several products exhibit fill rates below **55%**, indicating significant fulfillment challenges.
- These products should be prioritized for inventory replenishment, supplier review, and demand planning adjustments.
- Product-level analysis enables targeted improvements rather than broad inventory increases.

Business Decisions Supported

This dashboard enables operations and fulfillment managers to:

- Monitor customer service levels.
- Evaluate warehouse fulfillment performance.
- Compare regional demand against delivered quantity.
- Identify underperforming product categories.
- Prioritize low fill-rate products for replenishment.
- Reduce delayed orders.
- Improve inventory allocation.
- Enhance overall order fulfillment efficiency.

Executive Summary

The Order Fulfillment Dashboard provides a comprehensive view of customer demand fulfillment by integrating demand quantity, delivered quantity, fill rate, delayed orders, regional performance, warehouse efficiency, and product-level service metrics into a unified decision-support platform. SQL-validated Power BI measures enable operations teams to identify fulfillment gaps, monitor warehouse and product performance, reduce delivery delays, and optimize inventory allocation, ultimately improving customer service levels and overall supply chain responsiveness.

Page 7 – Warehouse Dashboard

Dashboard Objective

Monitor warehouse inventory performance by evaluating stock availability, inventory value, warehouse utilization, inventory health, and high-value SKUs to support inventory optimization and replenishment planning.

Business Questions Answered

- How much inventory is currently available?
- Which warehouse holds the highest inventory?
- Which warehouses contain the most at-risk inventory?
- Which product category contributes the highest inventory value?
- Which SKUs tie up the most inventory capital?
- What is the average inventory per warehouse?
- What is the average reorder point across all SKUs?

KPI Cards

---

1. Total Warehouses

Business Purpose

Displays the number of warehouses in the supply chain network.

DAX

```
Total Warehouses =
DISTINCTCOUNT(Inventory[Warehouse_ID])
```

SQL Validation

```
SELECT
COUNT(DISTINCT Warehouse_ID) AS Total_Warehouses
FROM Inventory;
```

---

2. Total Inventory

Business Purpose

Displays the total current inventory available across all warehouses.

DAX

```
Total Inventory =
SUM(Inventory[Current_Stock])
```

SQL Validation

```
SELECT
SUM(Current_Stock) AS Total_Inventory
FROM Inventory;
```

---

3. Avg Inventory / Warehouse

Business Purpose

Measures the average inventory held per warehouse.

DAX

```
Avg Inventory / WH =
DIVIDE(
    [Total Inventory],
    [Total Warehouses],
    0
)
```

SQL Validation

```
SELECT
SUM(Current_Stock) * 1.0 /
COUNT(DISTINCT Warehouse_ID) AS Avg_Inventory_Per_Warehouse
FROM Inventory;
```

---

4. Avg Reorder Point

Business Purpose

Displays the average inventory level at which replenishment should begin.

DAX

```
Avg Reorder Point =
AVERAGE(Inventory[Reorder_Point])
```

SQL Validation

```
SELECT
AVG(Reorder_Point) AS Avg_Reorder_Point
FROM Inventory;
```

---

5. At Risk SKUs

Business Purpose

Shows the number of unique SKUs currently below safety stock levels.

DAX

```
At Risk SKUs =
CALCULATE(
    DISTINCTCOUNT(Inventory[SKU_ID]),
    Inventory[Inventory Status] = "At Risk"
)
```

SQL Validation

```
SELECT
COUNT(DISTINCT SKU_ID) AS At_Risk_SKUs
FROM Inventory
WHERE Current_Stock < Safety_Stock;
```

---

6. Total Inventory Value

Business Purpose

Displays the monetary value of inventory currently held.

DAX

```
Total Inventory Value =
SUM(Inventory[Inventory_Value])
```

SQL Validation

```
SELECT
SUM(Inventory_Value) AS Total_Inventory_Value
FROM Inventory;
```

Visual 1 – Inventory by Warehouse

Objective

Compare total inventory across warehouses.

Business Purpose

Identifies which warehouses hold the largest inventory volumes.

DAX Used

Total Inventory

SQL Validation

```
SELECT
Warehouse_ID,
SUM(Current_Stock) AS Total_Inventory
FROM Inventory
GROUP BY Warehouse_ID
ORDER BY Total_Inventory DESC;
```

Visual 2 – Inventory Health by Warehouse

Objective

Compare healthy and at-risk SKUs across warehouses.

Business Purpose

Highlights warehouses requiring replenishment attention and those maintaining healthy stock levels.

DAX Used

At Risk SKUs

```
Healthy SKUs =
CALCULATE(
    DISTINCTCOUNT(Inventory[SKU_ID]),
    Inventory[Inventory Status] = "Healthy"
)
```

SQL Validation

```
SELECT
Warehouse_ID,
```

```
COUNT(DISTINCT CASE
WHEN Current_Stock < Safety_Stock
THEN SKU_ID
END) AS At_Risk_SKUs,

COUNT(DISTINCT CASE
WHEN Current_Stock >= Safety_Stock
THEN SKU_ID
END) AS Healthy_SKUs

FROM Inventory
GROUP BY Warehouse_ID
ORDER BY Warehouse_ID;
```

Visual 3 – Inventory Value by Category

Objective

Compare inventory investment across product categories.

Business Purpose

Identifies which categories consume the largest portion of inventory capital.

DAX Used

Total Inventory Value

SQL Validation

```
SELECT
    P.Category,
    SUM(I.Inventory_Value) AS Inventory_Value
FROM Inventory I
JOIN Product P
    ON I.SKU_ID = P.SKU_ID
GROUP BY P.Category
ORDER BY Inventory_Value DESC;
```

Visual 4 – Top 10 Inventory Value SKUs

Objective

Identify SKUs with the highest inventory value.

Business Purpose

Highlights products tying up the largest amount of working capital.

DAX Used

Total Inventory Value

SQL Validation

```
SELECT
    P.Product_Name,
    SUM(I.Inventory_Value) AS Inventory_Value
FROM Inventory I
JOIN Product P
    ON I.SKU_ID = P.SKU_ID
GROUP BY
    P.Product_Name
```

```
ORDER BY
    Inventory_Value DESC
LIMIT 10;
```

Skills Demonstrated

Power BI

- KPI Card Design
- Clustered Column Chart
- Stacked Bar Chart
- Horizontal Bar Chart
- Top N Ranking
- Interactive Slicers
- Conditional Formatting
- Dashboard Layout Design

DAX

- SUM()
- DISTINCTCOUNT()
- CALCULATE()
- DIVIDE()
- AVERAGE()
- Inventory Status Filtering
- Measure Reuse

SQL

- SUM()
- AVG()
- COUNT(DISTINCT)
- CASE WHEN
- GROUP BY
- ORDER BY
- LIMIT / TOP

Supply Chain Concepts

- Warehouse Performance
- Inventory Visibility
- Safety Stock Monitoring
- Reorder Planning
- Inventory Valuation
- Working Capital Management
- Inventory Health Analysis
- Warehouse Capacity Utilization

Key Business Findings

KPI Findings

- The organization manages inventory across **8 warehouses**, holding approximately **2M units** of stock.
- The average inventory per warehouse is **286K units**, indicating a relatively balanced inventory distribution.
- The average reorder point is **301 units**, serving as the replenishment trigger across SKUs.

- Approximately **234 SKUs** are currently below safety stock and require replenishment.
- Total inventory value is approximately **₹463M**, representing significant working capital invested in inventory.

Inventory by Warehouse

- **WH8** maintains the highest inventory level (**301K units**), while **WH6** holds the lowest (**255K units**).
- Inventory is fairly balanced across warehouses, suggesting effective stock distribution.

Inventory Health by Warehouse

- **WH7** contains the highest number of at-risk SKUs (**51**), indicating the greatest replenishment priority.
- **WH2** has the highest number of healthy SKUs (**261**), reflecting strong inventory availability.
- Most warehouses maintain substantially more healthy SKUs than at-risk SKUs, indicating overall stable inventory health.

Inventory Value by Category

- **Parts** account for the highest inventory investment (**₹142M**), followed by **Furniture (₹115M)**.
- **Appliances** contribute the lowest inventory value (**₹100M**).
- Inventory capital is concentrated in a few key product categories, highlighting opportunities for inventory optimization.

Top 10 Inventory Value SKUs

- The highest-value SKU contributes approximately **₹2.43M** in inventory value.
- Several SKUs individually exceed **₹2M**, indicating significant working capital concentration.
- These SKUs should be closely monitored to reduce excess inventory carrying costs and optimize stock turnover.

Business Decisions Supported

This dashboard enables warehouse and inventory managers to:

- Monitor inventory distribution across warehouses.
- Identify warehouses with the highest replenishment risk.
- Track inventory investment by product category.
- Monitor high-value SKUs consuming working capital.
- Prioritize replenishment based on safety stock.
- Improve warehouse inventory balancing.
- Optimize inventory carrying costs.
- Support strategic warehouse planning.

Executive Summary

The Warehouse Dashboard provides a comprehensive view of inventory availability, warehouse performance, inventory health, and inventory valuation across the supply chain network. By combining SQL-validated Power BI measures with warehouse-level and SKU-level analytics, the dashboard enables inventory planners to monitor stock distribution, prioritize replenishment activities, identify high-value inventory, and optimize working capital utilization while maintaining service levels.

Repository Structure

```
Supply-Chain-Control-Tower
│
├── Data
├── SQL
├── Power BI
├── Documentation
├── Images
├── README.md
└── LICENSE
```